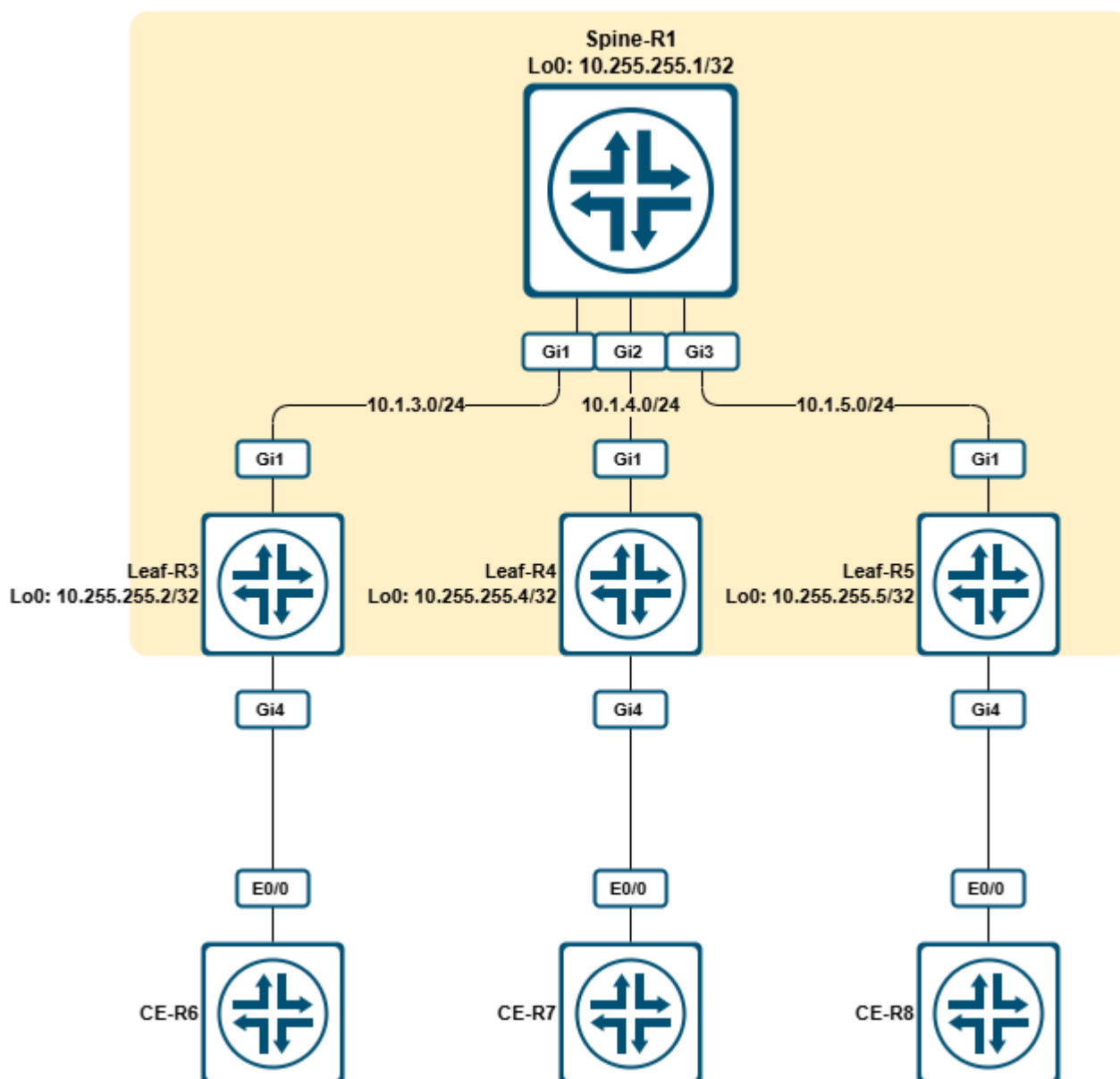


LAB - Cisco - VxLAN L2VNI Flood and learn

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Топология



Описание

В данной лабораторной работе необходимо организовать L2-связность между CE-маршрутизаторами. Для этого необходимо организовать VxLAN L2VNI в режиме Flood and learn.

Использованные образы:

- i86bi_LinuxL3-AdvEnterpriseK9-M2_157_3_May_2018.bin
- c8000v-17-13-01a

Настройка

Настройка - NVE-интерфейс

Исходим из того, что стартовая конфигурация (адресация, IGP, PIM) уже применена.

Проверяем RIB и наличие связности:

R1:

```
Spine-R1#show ip route | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
C       10.1.3.0/24 is directly connected, GigabitEthernet1
L       10.1.3.1/32 is directly connected, GigabitEthernet1
C       10.1.4.0/24 is directly connected, GigabitEthernet2
L       10.1.4.1/32 is directly connected, GigabitEthernet2
C       10.1.5.0/24 is directly connected, GigabitEthernet3
L       10.1.5.1/32 is directly connected, GigabitEthernet3
C       10.255.255.1/32 is directly connected, Loopback0
i L2    10.255.255.3/32 [115/10] via 10.1.3.3, 00:06:50, GigabitEthernet1
i L2    10.255.255.4/32 [115/10] via 10.1.4.4, 00:06:48, GigabitEthernet2
i L2    10.255.255.5/32 [115/10] via 10.1.5.5, 00:06:46, GigabitEthernet3
Spine-R1#tclsh
Spine-R1(tcl)#foreach IP {
+>10.255.255.3
+>10.255.255.4
+>10.255.255.5
+>} { ping $IP source Loopback0 }
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.255.3, timeout is 2 seconds:
Packet sent with a source address of 10.255.255.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.255.4, timeout is 2 seconds:
Packet sent with a source address of 10.255.255.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.255.5, timeout is 2 seconds:
Packet sent with a source address of 10.255.255.1
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Spine-R1(tcl)#
```

L3-связность, как видно, есть. Теперь проверим MRIB и значения RP:

R1:

```
Spine-R1#show ip pim rp mapping
Auto-RP is not enabled
PIM Group-to-RP Mappings

Group(s): 224.0.0.0/4, Static
  RP: 10.255.255.1 (?)
Spine-R1#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                          t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

IP Multicast Forwarding is not enabled.

Spine-R1#
```

R3:

```
Leaf-R3#show ip pim rp mapping
Auto-RP is not enabled
PIM Group-to-RP Mappings

Group(s): 224.0.0.0/4, Static
  RP: 10.255.255.1 (?)
Leaf-R3#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                          t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

Leaf-R3#
```

Как мы видим, в качестве RP настроен R1, MRIB пустой. В общем-то, все готово к настройке.

Для начала определимся с ролями маршрутизаторов: R1 – это транзитный маршрутизатор, который не будет на себе держать никакие сервисы (в терминологии SP – это P-маршрутизатор); R3, R4, R5 будут VTEP'ами (VxLAN Tunnel Endpoint – т.е. терминировать на себе VxLAN-туннели, в терминологии SP – PE-маршрутизаторами).

Чтобы настроить VTEP нам необходимы:

1. L3-связность между устройствами.

2. Работающий мультикаст (для репликации BUM-трафика через мультикаст)
3. Включенный и настроенный интерфейс NVE (network virtualization endpoint).

Первые два пункта у нас есть, создадим NVE-интерфейс на R3:

R3:

```
interface nve 1
  source-interface Loopback0
!
```

R3:

```
Leaf-R3(config)#interface nve 1
Leaf-R3(config-if)#
*May 18 2025 13:07:14.786 MSK: %LINK-3-UPDOWN: Interface nve1, changed state to
down
Leaf-R3(config-if)#source-interface Loopback0
*May 18 2025 13:07:36.841 MSK: %LINK-3-UPDOWN: Interface nve1, changed state to up
Leaf-R3(config-if)#
*May 18 2025 13:07:37.791 MSK: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Tunnel1, changed state to up
*May 18 2025 13:07:37.841 MSK: %LINEPROTO-5-UPDOWN: Line protocol on Interface
nve1, changed state to up
```

Что мы сделали? Мы создали интерфейс nve1, указали, что в качестве источника он должен использовать интерфейс Loopback0. После того, как мы указали src-интерфейс, nve1 поднялся, как и поднялся Tunnel1. Что это за интерфейс? А это технологический туннельный интерфейс для VxLAN-туннелей. Об этом нам говорят MUDP в качестве туннельного протокола и source_port == 4789 (UDP/4789 – это зарезервированный порт VxLAN).

R3:

```
Leaf-R3#show int tun1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  Interface is unnumbered. Using address of Loopback0 (10.255.255.3)
  MTU 9972 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 10.255.255.3
  Tunnel protocol/transport MUDP/IP
    TEID 0x0, sequencing disabled
    Checksumming of packets disabled
    source_port:4789, destination_port:0
  Tunnel TTL 255
  Tunnel transport MTU 1472 bytes
```

```
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)
Last input never, output never, output hang never
Last clearing of "show interface" counters 00:09:06
Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
  Received 0 broadcasts (0 IP multicasts)
  0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 packets output, 0 bytes, 0 underruns
  Output 0 broadcasts (0 IP multicasts)
  0 output errors, 0 collisions, 0 interface resets
  0 unknown protocol drops
  0 output buffer failures, 0 output buffers swapped out
Leaf-R3#
```

Эту же конфигурацию реплицируем на другие VTEP:

R4 / R5:

```
interface nve 1
  source-interface Loopback0
!
```

На данный момент у нас произведена первичная настройка 3 VTEP (R3 – R5) и все они готовы для разворачивания сервисов. Однако, прежде чем настраивать L2 VNI (VxLAN Network Id – в данном случае это обозначает отдельный vxlan-сервис), остановимся на том, как будет передаваться BUM-трафик.

BUM-трафик мы можем передавать несколькими путями:

1. Ingress replication
2. Multicast-group

Ingress replication – в данном случае BUM-трафик будет отправляться юникастом на каждого прописанного вручную VTEP'а внутри одного VNI. Т.е. на один вошедший на VTEP BUM-кадр, будет выпущено N VxLAN-пакетов, содержащих этот BUM-кадр, где N – это количество прописанных VTEP'ов. Не требует никаких дополнительных «вложений» – достаточно L3-связности между VTEP'ами.

Multicast-group – в данном случае BUM-трафик будет распространяться в рамках многоадресной рассылки. Очевидно, требует развернутой multicast-сети.

В рамках данной лабораторной работы мы настроим два L2 VNI:

1. 10001 – VNI с Ingress Replication
2. 10002 – VNI с Multicast-group

Настройка - VNI 10001 – L2VNI Ingress Replication

Начнем с VNI 10001:

R3:

```
Leaf-R3#debug nve all
NVE Manager all debugging is on
Leaf-R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Leaf-R3(config)#int nve 1
Leaf-R3(config-if)#member vni 10001
Leaf-R3(config-if-nve-vni)#ingress-replication 10.255.255.4
Leaf-R3(config-if-nve-vni)#
*May 18 2025 14:00:51.691 MSK: NVE-MGR-DB: [IF 0xE]VNI node creation
*May 18 2025 14:00:51.691 MSK: NVE-MGR-DB: Allocate VNI ID 10001
*May 18 2025 14:00:51.691 MSK: NVE-MGR-DB: Lock VNI ID 10001
*May 18 2025 14:00:51.691 MSK: NVE-MGR-DB: VNI Node created [77349A4AA910]
*May 18 2025 14:00:51.691 MSK: NVE-MGR-DB: Insert individual VNI 10001 into nveos-
>vni_list
*May 18 2025 14:00:51.692 MSK: NVE-MGR-PARSER: VNI bitlist: 10001
*May 18 2025 14:00:51.692 MSK: NVE-MGR-PD: VNI 10001 create notification to PD
*May 18 2025 14:00:51.695 MSK: NVE-MGR-PD: VNI 10001 Create notif successful, map
[pd 0x404011] to [pi 0x77349A4AA910]
*May 18 2025 14:00:51.695 MSK: NVE-MGR-DB: creating peer node for 10.255.255.4
*May 18 2025 14:00:51.695 MSK: NVE-DB-EVT: Add peer 10.255.255.4 on VNI 10001 on
interface 0xE
*May 18 2025 14:00:51.695 MSK: NVE-DB-EVT: Added peer 10.255.255.4 on VNI 10001 on
NVE 1
*May 18 2025 14:00:51.695 MSK: NVE-DB-EVT: Add VNI 10001 to peer 10.255.255.4
source 4
*May 18 2025 14:00:51.695 MSK: NVE-DB-EVT: Add VNI 10001 to peer 10.255.255.4 NVE
1 rc 0 flags 4 state 32
*May 18 2025 14:00:51.695 MSK: NVE-DB-EVT: Added VNI 10001 on Peer 10.255.255.4
*May 18 2025 14:00:51.697 MSK: NVE-MGR-TUNNEL: Tunnel Endpoint 10.255.255.4 added,
dport 4789
*May 18 2025 14:00:51.697 MSK: NVE-MGR-EI: Notify L2FIB of install addr
10.255.255.4 for vni 10001
*May 18 2025 14:00:51.697 MSK: NVE-MGR-TUNNEL: Endpoint 10.255.255.4 already added
*May 18 2025 14:00:51.697 MSK: NVE-MGR-EI: Notifying BD engine of VNI 10001 create
*May 18 2025 14:00:51.697 MSK: NVE-MGR-STATE: VNI 10001 NVE if is up
*May 18 2025 14:00:51.697 MSK: NVE-MGR-STATE: VNI 10001 BD unbinded or down
*May 18 2025 14:00:51.697 MSK: NVE-MGR-STATE: VNI 10001 down reason: BD is down
*May 18 2025 14:00:51.697 MSK: NVE-MGR-STATE: vni 10001: Notify clients of state
change Create to BD Down/Removed, dirty 0
Leaf-R3(config-if-nve-vni)#
*May 18 2025 14:00:51.698 MSK: NVE-MGR-PD: VNI 10001 Create to BD Down/Removed
State update to PD successful
*May 18 2025 14:00:51.698 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 14:00:51.698 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 14:00:51.698 MSK: NVE-MGR-EI: No BD association for vni 10001
*May 18 2025 14:00:51.698 MSK: NVE-MGR-STATE: vni 10001: New State as a result of
create BD Down/Removed
*May 18 2025 14:00:51.698 MSK: NVE-MGR-PARSER: VNI 10001 ingr-repl 10.255.255.4
created
Leaf-R3(config-if-nve-vni)#ingress-replication 10.255.255.5
```

```

Leaf-R3(config-if-nve-vni)#
*May 18 2025 14:01:02.036 MSK: NVE-MGR-DB: [IF 0xE]VNI node creation
*May 18 2025 14:01:02.036 MSK: NVE-MGR-DB: VNI update for 10001, local_routing 0
*May 18 2025 14:01:02.036 MSK: NVE-MGR-DB: creating peer node for 10.255.255.5
*May 18 2025 14:01:02.036 MSK: NVE-DB-EVT: Add peer 10.255.255.5 on VNI 10001 on
interface 0xE
*May 18 2025 14:01:02.036 MSK: NVE-DB-EVT: Added peer 10.255.255.5 on VNI 10001 on
NVE 1
*May 18 2025 14:01:02.036 MSK: NVE-DB-EVT: Add VNI 10001 to peer 10.255.255.5
source 4
*May 18 2025 14:01:02.036 MSK: NVE-DB-EVT: Add VNI 10001 to peer 10.255.255.5 NVE
1 rc 0 flags 4 state 32
Leaf-R3(config-if-nve-vni)#
*May 18 2025 14:01:02.036 MSK: NVE-DB-EVT: Added VNI 10001 on Peer 10.255.255.5
*May 18 2025 14:01:02.037 MSK: NVE-MGR-TUNNEL: Tunnel Endpoint 10.255.255.5 added,
dport 4789
*May 18 2025 14:01:02.037 MSK: NVE-MGR-EI: Notify L2FIB of install addr
10.255.255.5 for vni 10001
*May 18 2025 14:01:02.037 MSK: NVE-MGR-PARSER: VNI 10001 ingr-repl 10.255.255.5
created
Leaf-R3(config-if-nve-vni)#end
Leaf-R3#show
*May 18 2025 14:01:12.510 MSK: %SYS-5-CONFIG_I: Configured from console by console
Leaf-R3#show run int nve 1
Building configuration...

```

Current configuration : 186 bytes

```

!
interface nve1
 no ip address
 source-interface Loopback0
 member vni 10001
  ingress-replication 10.255.255.4
  ingress-replication 10.255.255.5
!
 no mop enabled
 no mop sysid
end

```

Leaf-R3#show nve vni 10001

Interface	VNI	Multicast-group	VNI state	Mode	BD	cfg	vrf
nve1	10001	N/A	BD Down/Re	L2DP	N/A	CLI	N/A

Leaf-R3#

*May 18 2025 14:01:31.749 MSK: NVE-MGR-PARSER: show nve vni 10001

Leaf-R3#show nve vni 10001 detail

Interface	VNI	Multicast-group	VNI state	Mode	BD	cfg	vrf
nve1	10001	N/A	BD Down/Re	L2DP	N/A	CLI	N/A

VNI down reason:

BD un-configured or down

VNI Detailed statistics:

Pkts In	Bytes In	Pkts Out	Bytes Out
0	0	0	0

Leaf-R3#show nve peers

'M' - MAC entry download flag 'A' - Adjacency download flag
 '4' - IPv4 flag '6' - IPv6 flag

Interface	VNI	Type	Peer-IP	RMAC/Num_RTs	eVNI	state	flags	UP
nve1	10001	L2DP	10.255.255.4	----	-	--	-/-	
nve1	10001	L2DP	10.255.255.5	----	-	--	-/-	

Что здесь произошло? Сначала мы включили дебаг всего для NVE. После этого мы зашли в настройки интерфейса nve1 и создали новый VNI (member vni 10001). Внутри этого VNI мы включили ingress-replication для 10.255.255.4 и 10.255.255.5.

Теперь настроим подключение клиента. Для этого сначала настроим EFP (Ethernet Flow Point):

R3:

```
interface GigabitEthernet4
description NE=CE-R6_Eth0/0
service instance 6 ethernet
encapsulation dot1q 6
rewrite ingress tag pop 1 symmetric
!
```

Здесь:

- **service-instance 6 ethernet** – это создание EFP под номером 6.
- **encapsulation dot1q 6** – указываем, что данный EFP должен обрабатывать dot1q-кадры с VID == 6.
- **rewrite ingress tag pop 1 symmetric** – мы указываем, что для входящих (ingress) кадров, попадающих в EFP, мы изменяем (rewrite) метку, а именно удаляем 1 внешнюю метку (pop 1). Symmetric означает, что для исходящих из EFP кадров мы проводим обратную манипуляцию (т.е. добавляем внешний dot1q-заголовок с VID == 6).

Теперь свяжем EFP и VNI через bridge-domain:

R3:

```
bridge-domain 1001
member Gi4 service-instance 6
member vni 10001
!
```

R3:

```
Leaf-R3(config)#bridge-domain 1001
Leaf-R3(config-bdmain)#member Gi4 service-instance 6
Leaf-R3(config-bdmain-efp)#exit
Leaf-R3(config-bdmain)#member vni 10001
```



```
Leaf-R3(config-bdomain)#
*May 18 2025 14:09:03.213 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
*May 18 2025 14:09:03.214 MSK: NVE-MGR-DB: Return vni state BD Down/Removed for
pi_hdl[0x77349A4AA910], vni[10001]
*May 18 2025 14:09:03.214 MSK: NVE-MGR-DB: Return vni state BD Down/Removed for
pi_hdl[0x77349A4AA910], vni[10001]
*May 18 2025 14:09:03.214 MSK: NVE-MGR-EI: L2FIB query info for BD 0, VNI 10001
*May 18 2025 14:09:03.215 MSK: NVE-MGR-EI: PP up notification for bd_id 1001
*May 18 2025 14:09:03.215 MSK: NVE-DB-EVT: Initialized BD(1001) node for VNI 10001
*May 18 2025 14:09:03.215 MSK: NVE-MGR-EI: DP VNI 10001 doesn't support on demand
mcast join
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: VNI 10001 NVE if is up
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: VNI 10001 BD 1001 binded and UP
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: VNI 10001 state_flags 0xA
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: vni 10001: Notify clients of state
change BD Down/Removed to Up, dirty 0
*May 18 2025 14:09:03.215 MSK: NVE-MGR-PD: VNI 10001 BD Down/Removed to Up State
update to PD successful
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 14:09:03.215 MSK: NVE-MGR-EI: Notify L2FIB of updated tunnel handle
0xF for vni 10001
*May 18 2025 14:09:03.215 MSK: NVE-MGR-EI: DP VNI 10001 doesn't support on demand
mcast join
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: VNI 10001 NVE if is up
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: VNI 10001 BD 1001 binded and UP
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: VNI 10001 state_flags 0xA
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: vni 10001: Notify clients of state
change Up to Up, dirty 0
*May 18 2025 14:09:03.215 MSK: NVE-MGR-PD: VNI 10001 Up to Up State update to PD
successful
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 14:09:03.215 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 14:09:03.215 MSK: NVE-MGR-EI: Notify L2FIB of updated tunnel handle
0xF for vni 10001
*May 18 2025 14:09:03.216 MSK: NVE-MGR-EI: VNI 10001: BD state changed to up, vni
state to Up
*May 18 2025 14:09:03.234 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
*May 18 2025 14:09:03.249 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
*May 18 2025 14:09:03.249 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
*May 18 2025 14:09:03.249 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
*May 18 2025 14:09:03.249 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
Leaf-R3(config-bdomain)#
*May 18 2025 14:09:03.249 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
*May 18 2025 14:09:03.249 MSK: NVE-MGR-DB: Return pi_hdl[0x77349A4AA910] for vni
10001
Leaf-R3(config-bdomain)#end
```

В дебаге мы видим, что к VNI 10001 был привязан bridge-domain 1001, сам VNI перешел в состояние UP. Посмотрим, что видно в BD:

R3:

```
Leaf-R3#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 6
    vni 10001
    AED MAC address      Policy Tag      Age Pseudoport
    -----
```

На данный момент таблица MAC-адресов пуста, т.к. у нас ничего нет со стороны клиента. Настроим на CE-R6:

R6:

```
interface Ethernet0/0.6
 encapsulation dot1q 6
 description VNI_10001
 ip address 10.10.1.6 255.255.255.0
!
```

После того, как мы введем IP-адрес, R6 отправит Gratuitous ARP. Но перед этим стоит включить дампы трафика на R3, Gi1:

2369	2025-05-18 14:37:37,101477	50:00:00:01:00:00	09:00:2b:00:00:05	ISIS H.	1513 P2P HELLO, System-ID: 0102.5525.5001
2370	2025-05-18 14:37:38,930190	50:00:00:01:00:00	09:00:2b:00:00:04	ISIS	60 IS HELLO
2371	2025-05-18 14:37:42,086927	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2372	2025-05-18 14:37:42,086937	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2373	2025-05-18 14:37:42,089382	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2374	2025-05-18 14:37:42,089390	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2375	2025-05-18 14:37:45,075639	50:00:00:03:00:00	09:00:2b:00:00:05	ISIS H.	1513 P2P HELLO, System-ID: 0102.5525.5003
2376	2025-05-18 14:37:45,836607	50:00:00:01:00:00	09:00:2b:00:00:05	ISIS H.	1513 P2P HELLO, System-ID: 0102.5525.5001


```
> Frame 2371: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface -, id 0
> Ethernet II, Src: 50:00:00:03:00:00, Dst: 50:00:00:01:00:00
> Internet Protocol Version 4, Src: 10.255.255.3, Dst: 10.255.255.4
> User Datagram Protocol, Src Port: 57177, Dst Port: 4789
> Virtual eXtensible Local Area Network
> Ethernet II, Src: aa:bb:cc:00:60:00, Dst: ff:ff:ff:ff:ff:ff
> Address Resolution Protocol (reply/gratuitous ARP)
```

2370	2025-05-18 14:37:38,930190	50:00:00:01:00:00	09:00:2b:00:00:04	ISIS	60 IS HELLO
2371	2025-05-18 14:37:42,086927	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2372	2025-05-18 14:37:42,086937	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2373	2025-05-18 14:37:42,089382	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2374	2025-05-18 14:37:42,089390	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110 Gratuitous ARP for 10.10.1.6 (Reply)
2375	2025-05-18 14:37:45,075639	50:00:00:03:00:00	09:00:2b:00:00:05	ISIS H.	1513 P2P HELLO, System-ID: 0102.5525.5003
2376	2025-05-18 14:37:45,836607	50:00:00:01:00:00	09:00:2b:00:00:05	ISIS H.	1513 P2P HELLO, System-ID: 0102.5525.5001

```

> Frame 2372: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface -, id 0
> Ethernet II, Src: 50:00:00:03:00:00, Dst: 50:00:00:01:00:00
> Internet Protocol Version 4, Src: 10.255.255.3, Dst: 10.255.255.5
> User Datagram Protocol, Src Port: 57177, Dst Port: 4789
> Virtual eXtensible Local Area Network
> Ethernet II, Src: aa:bb:cc:00:60:00, Dst: ff:ff:ff:ff:ff:ff
> Address Resolution Protocol (reply/gratuitous ARP)

```

На скринах мы видим, что GrARP был отправлен юникастовыми пакетами с 10.255.255.3 (Lo0 R3) на 10.255.255.4 и 10.255.255.5. При этом стоит обратить внимание на то, что внутри VxLAN-пакета находится Ethernet-кадр без dot1q-заголовка.

Посмотрим, что у нас на R3:

R3:

```

Leaf-R3#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 6
    vni 10001
    AED MAC address      Policy Tag      Age Pseudoport
    -----
    0  AABB.CC00.6000 forward dynamic  15  GigabitEthernet4.EFP6

Leaf-R3#show nve vni 10001 detail
Interface VNI      Multicast-group VNI state Mode BD   cfg vrf
nve1      10001      N/A           Up      L2DP 1001 CLI N/A
VNI Detailed statistics:
    Pkts In  Bytes In  Pkts Out  Bytes Out
    0         0         4         272

Leaf-R3#

```

BD изучил MAC-адрес R6 за Gi4.EFP6. А в статистике VNI счетчик отправленных пакетов увеличился на 4.

Теперь подключим и CE-R7. Для этого на R4 делаем аналогичные действия:

Создаем VNI:

R4:

```

interface nve1
member vni 10001
ingress-replication 10.255.255.3

```

```
ingress-replication 10.255.255.5
!
```

Создаем EFP:

R4:

```
interface GigabitEthernet4
description NE=CE-R7_Eth0/0
service instance 7 ethernet
encapsulation dot1q 7
rewrite ingress tag pop 1 symmetric
!
```

Связываем EFP и VNI через bridge-domain:

R4:

```
bridge-domain 1001
member Gi4 service-instance 7
member vni 10001
!
```

Проверяем:

R4:

```
Leaf-R4#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 7
    vni 10001
    AED MAC address      Policy  Tag      Age  Pseudoport
    -----
Leaf-R4#
```

Теперь настроим CE-R7:

R7:

```
interface Ethernet0/0.7
encapsulation dot1q 7
```

```
ip address 10.10.1.7 255.255.255.0
!
```

Также, как и R6, R7 отправил GratARP, который дошел до R3 и до R6

vxlan						
No.	Time	Source	Destination	Protocol	Length	Info
2371	2025-05-18 14:37:42,086927	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2372	2025-05-18 14:37:42,086937	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2373	2025-05-18 14:37:42,089382	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2374	2025-05-18 14:37:42,089390	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2923	2025-05-18 14:58:41,209388	aa:bb:cc:00:70:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.7 (Reply)
2924	2025-05-18 14:58:41,212304	aa:bb:cc:00:70:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.7 (Reply)

```
> Frame 2923: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface -, id 0
> Ethernet II, Src: 50:00:00:01:00:00, Dst: 50:00:00:03:00:00
> Internet Protocol Version 4, Src: 10.255.255.4, Dst: 10.255.255.3
> User Datagram Protocol, Src Port: 57765, Dst Port: 4789
> Virtual eXtensible Local Area Network
  > Flags: 0x0000, VXLAN Network ID (VNI)
    Group Policy ID: 0
    VXLAN Network Identifier (VNI): 10001
    Reserved: 0
> Ethernet II, Src: aa:bb:cc:00:70:00, Dst: ff:ff:ff:ff:ff:ff
> Address Resolution Protocol (reply/gratuitous ARP)
```

R3:

```
Leaf-R3#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
  GigabitEthernet4 service instance 6
  vni 10001
  AED MAC address      Policy Tag      Age Pseudoport
  -----
  0 AABB.CC00.6000 forward dynamic 294 GigabitEthernet4.EFP6
  0 AABB.CC00.7000 forward dynamic 293 nve1.VNI10001, VxLAN
                                     src: 10.255.255.3 dst: 10.255.255.4

Leaf-R3#
```

R4:

```
CE-R6#ping 10.10.1.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.1.7, timeout is 2 seconds:
!!!!
```

```
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
CE-R6#
```

На R4 BD выглядит так:

R4:

```
Leaf-R4#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 7
        vni 10001
            AED MAC address      Policy Tag      Age Pseudoport
            -----
            0 AABB.CC00.6000 forward dynamic 270 nve1.VNI10001, VxLAN
                                   src: 10.255.255.4 dst: 10.255.255.3
            0 AABB.CC00.7000 forward dynamic 270 GigabitEthernet4.EFP7

Leaf-R4#
```

При этом стоит обратить внимание:

No.	Time	Source	Destination	Protocol	Length	Info
2371	2025-05-18 14:37:42,006927	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2372	2025-05-18 14:37:42,006937	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2373	2025-05-18 14:37:42,009382	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2374	2025-05-18 14:37:42,009390	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.6 (Reply)
2923	2025-05-18 14:58:41,209388	aa:bb:cc:00:70:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.7 (Reply)
2924	2025-05-18 14:58:41,212304	aa:bb:cc:00:70:00	ff:ff:ff:ff:ff:ff	ARP	110	Gratuitous ARP for 10.10.1.7 (Reply)
2982	2025-05-18 15:00:47,061404	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Who has 10.10.1.7? Tell 10.10.1.6
2983	2025-05-18 15:00:47,061415	aa:bb:cc:00:60:00	ff:ff:ff:ff:ff:ff	ARP	110	Who has 10.10.1.7? Tell 10.10.1.6
2984	2025-05-18 15:00:47,062774	aa:bb:cc:00:70:00	aa:bb:cc:00:60:00	ARP	110	10.10.1.7 is at aa:bb:cc:00:70:00
2985	2025-05-18 15:00:49,063388	10.10.1.6	10.10.1.7	ICMP	164	Echo (ping) request id=0x0001, seq=1/256, ttl=255 (reply in 2986)
2986	2025-05-18 15:00:49,064148	10.10.1.7	10.10.1.6	ICMP	164	Echo (ping) reply id=0x0001, seq=1/256, ttl=255 (request in 2985)
2987	2025-05-18 15:00:49,064549	10.10.1.6	10.10.1.7	ICMP	164	Echo (ping) request id=0x0001, seq=2/512, ttl=255 (reply in 2988)
2988	2025-05-18 15:00:49,064931	10.10.1.7	10.10.1.6	ICMP	164	Echo (ping) reply id=0x0001, seq=2/512, ttl=255 (request in 2987)
2989	2025-05-18 15:00:49,065123	10.10.1.6	10.10.1.7	ICMP	164	Echo (ping) request id=0x0001, seq=3/768, ttl=255 (reply in 2990)
2990	2025-05-18 15:00:49,065406	10.10.1.7	10.10.1.6	ICMP	164	Echo (ping) reply id=0x0001, seq=3/768, ttl=255 (request in 2989)
2991	2025-05-18 15:00:49,065663	10.10.1.6	10.10.1.7	ICMP	164	Echo (ping) request id=0x0001, seq=4/1024, ttl=255 (reply in 2992)
2992	2025-05-18 15:00:49,066137	10.10.1.7	10.10.1.6	ICMP	164	Echo (ping) reply id=0x0001, seq=4/1024, ttl=255 (request in 2991)


```
> Frame 2985: 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface -, id 0
> Ethernet II, Src: 50:00:00:03:00:00, Dst: 50:00:00:01:00:00
> Internet Protocol Version 4, Src: 10.255.255.3, Dst: 10.255.255.4
> User Datagram Protocol, Src Port: 60581, Dst Port: 4789
> Virtual eXtensible Local Area Network
  > Flags: 0x0000, VXLAN Network ID (VNI)
    Group Policy ID: 0
    VXLAN Network Identifier (VNI): 10001
    Reserved: 0
> Ethernet II, Src: aa:bb:cc:00:60:00, Dst: aa:bb:cc:00:70:00
> Internet Protocol Version 4, Src: 10.10.1.6, Dst: 10.10.1.7
> Internet Control Message Protocol
```

Мы явно видим, что ICMP-запросы шли только между R3 и R4, т.к. это уже known-unicast.

Статистика с R3:

R3:

```
Leaf-R3#show nve vni 10001 detail
Interface VNI      Multicast-group VNI state Mode BD   cfg vrf
nve1      10001          N/A           Up      L2DP  1001 CLI N/A
```

VNI Detailed statistics:

	Pkts In	Bytes In	Pkts Out	Bytes Out
	7	692	10	896

Leaf-R3#

Также стоит обратить внимание на следующее: казалось бы, R6 и R7 находятся в разных вланах (6 и 7 соответственно), но, при этом, связность вполне себе есть. Дело в том, что в рамках EFP мы явно указали, что внешний тег должен быть снят при попадании в EFP (а на выходе наоборот, добавить), что мы видели в дампах, поэтому никаких противоречий здесь нет.

Включим OSPF между R6 и R7:

R6:

```
interface Ethernet0/0.6
ip ospf 10001 area 0
```

R7:

```
interface Ethernet0/0.7
ip ospf 10001 area 0
```

Проверяем:

R6:

```
CE-R6#show ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.1.7	1	FULL/DR	00:00:39	10.10.1.7	Ethernet0/0.6

CE-R6#

R7:

```
CE-R7#show ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.1.6	1	FULL/BDR	00:00:34	10.10.1.6	Ethernet0/0.7

CE-R7#

А также на R8 осуществим аналогичные настройки (создадим интерфейс и включим OSPF):

R8:

```

interface GigabitEthernet1
  service instance 8 ethernet
  encapsulation dot1q 8
  exit
exit
bridge-domain 8
  member GigabitEthernet1 service-instance 8
exit
interface BDI8
  ip address 10.10.1.8 255.255.255.0
  no shutdown
  ip ospf 10001 area 0
!
```

Примечание: настройка на CE-R8 выглядит нестандартно из-за проблем с вложенной эмуляцией сетевых адаптеров (особенно vmxnet) в EVE-NG CE 6.x, из-за чего для тегированного трафика нормально работает только такой вариант в паре с virtio-net-pci (актуально в первую очередь для CSR1kv и C8kv).

На R5 настроим сервис:

R5:

```

interface GigabitEthernet4
  service instance 8 ethernet
  encapsulation dot1q 8
  rewrite ingress tag pop 1 symmetric
  exit
exit
interface nve1
  member vni 10001
  ingress-replication 10.255.255.3
  ingress-replication 10.255.255.4
!
bridge-domain 1001
  member vni 10001
  member GigabitEthernet4 service-instance 8
```

Проверяем:

R6:

```
CE-R6#show ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.1.7	1	FULL/DR	00:00:35	10.10.1.7	Ethernet0/0.6
10.10.1.8	1	FULL/DROTHER	00:00:33	10.10.1.8	Ethernet0/0.6

```
CE-R6#
```


R8:

```
CE-R8#show ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.1.6	1	FULL/BDR	00:00:31	10.10.1.6	BDI8
10.10.1.7	1	FULL/DR	00:00:39	10.10.1.7	BDI8

```
CE-R8#
```

Для проверки добавим на каждый CE Loopback101 с IP-адресами 101.0.0.\$R_NUM/32 и добавим их в OSPF:

R6:

```
interface Loopback101
 ip address 101.0.0.6 255.255.255.255
 ip ospf 10001 area 0
!
```

R7:

```
interface Loopback101
 ip address 101.0.0.7 255.255.255.255
 ip ospf 10001 area 0
!
```

R8:

```
interface Loopback101
 ip address 101.0.0.8 255.255.255.255
 ip ospf 10001 area 0
!
```

Проверим:

R6:

```
CE-R6#show ip route ospf | b Gate
Gateway of last resort is not set

    101.0.0.0/32 is subnetted, 3 subnets
O        101.0.0.7 [110/11] via 10.10.1.7, 00:00:21, Ethernet0/0.6
O        101.0.0.8 [110/11] via 10.10.1.8, 00:00:21, Ethernet0/0.6
CE-R6#tclsh
CE-R6(tcl)#foreach IP {
```

```
+>101.0.0.7
+>101.0.0.8
+>} { ping $IP source Loopback101 }
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 101.0.0.7, timeout is 2 seconds:
Packet sent with a source address of 101.0.0.6
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 101.0.0.8, timeout is 2 seconds:
Packet sent with a source address of 101.0.0.6
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
CE-R6(tcl)#
```

R7:

```
CE-R7#show ip route ospf | b Gate
Gateway of last resort is not set

    101.0.0.0/32 is subnetted, 3 subnets
O        101.0.0.6 [110/11] via 10.10.1.6, 00:00:40, Ethernet0/0.7
O        101.0.0.8 [110/11] via 10.10.1.8, 00:00:40, Ethernet0/0.7
CE-R7#tclsh
CE-R7(tcl)#foreach IP {
+>101.0.0.6
+>101.0.0.8
+>} { ping $IP source Loopback101 }
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 101.0.0.6, timeout is 2 seconds:
Packet sent with a source address of 101.0.0.7
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 101.0.0.8, timeout is 2 seconds:
Packet sent with a source address of 101.0.0.7
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
CE-R7(tcl)#
```

R8:

```
CE-R8#show ip route ospf | b Gate
Gateway of last resort is not set

    101.0.0.0/32 is subnetted, 3 subnets
O        101.0.0.6 [110/2] via 10.10.1.6, 00:01:04, BDI8
O        101.0.0.7 [110/2] via 10.10.1.7, 00:01:08, BDI8
CE-R8#tclsh
CE-R8(tcl)#foreach IP {
```

```

+>101.0.0.6
+>101.0.0.7
+>} { ping $IP source Loopback101 }
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 101.0.0.6, timeout is 2 seconds:
Packet sent with a source address of 101.0.0.8
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 101.0.0.7, timeout is 2 seconds:
Packet sent with a source address of 101.0.0.8
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
CE-R8(tcl)#

```

Соответственно, мы видим, что связность есть между всеми CE-устройствами. На VTEP'ах в бриджах видим такое:

R3:

```

Leaf-R3#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 6
    vni 10001
    AED MAC address      Policy Tag      Age Pseudoport
    -----
    0  AABB.CC00.6000 forward dynamic  298 GigabitEthernet4.EFP6
    0  001E.F69F.AEBF forward dynamic  300 nve1.VNI10001, VxLAN
                                   src: 10.255.255.3 dst: 10.255.255.5
    0  AABB.CC00.7000 forward dynamic  298 nve1.VNI10001, VxLAN
                                   src: 10.255.255.3 dst: 10.255.255.4

```

R4:

```

Leaf-R4#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 7
    vni 10001
    AED MAC address      Policy Tag      Age Pseudoport
    -----
    0  AABB.CC00.6000 forward dynamic  293 nve1.VNI10001, VxLAN
                                   src: 10.255.255.4 dst: 10.255.255.3

```

```

0    001E.F69F.AEBF forward dynamic    294    nve1.VNI10001, VxLAN
                                     src: 10.255.255.4 dst: 10.255.255.5
0    AABB.CC00.7000 forward dynamic    292    GigabitEthernet4.EFP7

```

R5:

```

Leaf-R5#show bridge-domain 1001
Bridge-domain 1001 (2 ports in all)
State: UP                               Mac learning: Enabled
Aging-Timer: 300 second(s)
Unknown Unicast Flooding Suppression: Disabled
Maximum address limit: 65536
    GigabitEthernet4 service instance 8
    vni 10001
    AED MAC address      Policy Tag      Age Pseudoport
    -----
0    AABB.CC00.6000 forward dynamic    298    nve1.VNI10001, VxLAN
                                     src: 10.255.255.5 dst: 10.255.255.3
0    001E.F69F.AEBF forward dynamic    291    GigabitEthernet4.EFP8
0    AABB.CC00.7000 forward dynamic    297    nve1.VNI10001, VxLAN
                                     src: 10.255.255.5 dst: 10.255.255.4

```

На этом создание L2VNI в режиме Flood and Learn, с Ingress replication для BUM-трафика завершено.

Настройка - VNI 10002 – L2VNI Multicast group

Теперь перейдем к настройке L2VNI с multicast-group для рассылки BUM-трафика. В рамках этого задания будем использовать следующее:

- Номер VNI: 10002.
- Номер VLAN (стык VTEP <> CE): 678 (при этом, dot1q-метку мы должны будем оставить внутри VxLAN-пакета).
- Адрес группы: 239.6.7.8.

Для начала посмотрим, что у нас в MRIB на каждом роутере:

R1:

```

Spine-R1#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

IP Multicast Forwarding is not enabled.

```

R3:

```
Leaf-R3#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode
```

R4:

```
Leaf-R4#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode
```

R5:

```
Leaf-R5#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode
```

MRIB у нас пустой.

Для начала создадим VNI на VTEP R3:

R3:

```
interface nve1
 member vni 10002
 mcast-group 239.6.7.8
!
```

R3:

```
Leaf-R3#debug nve all
NVE Manager all debugging is on
Leaf-R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Leaf-R3(config)#interface nve1
Leaf-R3(config-if)#member vni 10002
Leaf-R3(config-if-nve-vni)#mcast-group 239.6.7.8
Leaf-R3(config-if-nve-vni)#
*May 18 2025 16:27:57.986 MSK: NVE-MGR-DB: [IF 0xE]VNI node creation
*May 18 2025 16:27:57.986 MSK: NVE-MGR-DB: Allocate VNI ID 10002
```

```

*May 18 2025 16:27:57.986 MSK: NVE-MGR-DB: Lock VNI ID 10002
*May 18 2025 16:27:57.986 MSK: NVE-MGR-DB: VNI Node created [77349A4AA7B8]
*May 18 2025 16:27:57.986 MSK: NVE-MGR-DB: Insert individual VNI 10002 into nveos-
>vni_list
*May 18 2025 16:27:57.986 MSK: NVE-MGR-PARSER: VNI bitlist: 10001-10002
*May 18 2025 16:27:57.986 MSK: NVE-MGR-PD: VNI 10002 create notification to PD
*May 18 2025 16:27:58.000 MSK: NVE-MGR-PD: VNI 10002 Create notif successful, map
[pd 0x404013] to [pi 0x77349A4AA7B8]
*May 18 2025 16:27:58.000 MSK: NVE-MGR-MCAST: creating mcast node for 239.6.7.8
*May 18 2025 16:27:58.009 MSK: NVE-MGR-MCAST: IGMP add for (0.0.0.0,239.6.7.8) was
successful
*May 18 2025 16:27:58.018 MSK: NVE-MGR-TUNNEL: Tunnel Endpoint 239.6.7.8 added,
dport 4789
*May 18 2025 16:27:58.018 MSK: NVE-MGR-TUNNEL: Endpoint 239.6.7.8 added
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: Mcast processed vni 10002 create on
bay 0
*May 18 2025 16:27:58.018 MSK: NVE-MGR-EI: Notifying BD engine of VNI 10002 create
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: VNI 10002 NVE if is up
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: VNI 10002 BD unbinded or down
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: VNI 10002 down reason: BD is down
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: vni 10002: Notify clients of state
change Create to BD Down/Removed, dirty 0
*May 18 2025 16:27:58.018 MSK: NVE-MGR-PD: VNI 10002 Create to BD Down/Removed
State update to PD successful
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: No Mcast Node found
*May 18 2025 16:27:58.018 MSK: NVE-MGR-EI: No BD association for vni 10002
*May 18 2025 16:27:58.018 MSK: NVE-MGR-STATE: vni 10002: New State as a result of
create BD Down/Removed
*May 18 2025 16:27:58.018 MSK: NVE-MGR-PARSER: VNI 10002 mcast 239.6.7.8 UNKNOWN
created, lrouting 0
*May 18 2025 16:27:59.481 MSK: %PIM-5-DRCHG: DR change from neighbor 0.0.0.0 to
10.255.255.3 on interface Tunnel1
Leaf-R3(config-if-nve-vni)#end

```

Итак, что мы здесь видим? А видим мы то, что создали VNI 10002 с группой 239.6.7.8, из-за чего была добавлена подписка на эту группу. Также был создан endpoint udp/239.6.7.8:4789. Сам по себе VNI ушел в down, т.к. пока не привязан к bridge-domain. И в конце был включен PIM на интерфейсе Tunnel1 (напоминаю, это туннельный интерфейс MUDP/IP для NVE).

На всякий случай проверим вышесказанное и, заодно, посмотрим MRIB:

R3:

```

*May 18 2025 16:28:12.629 MSK: %SYS-5-CONFIG_I: Configured from console by console
Leaf-R3#show ip pim interface

```

Address	Interface	Ver/ Mode	Nbr Count	Query Intvl	DR Prior	DR
10.255.255.3	Loopback0	v2/S	0	30	1	
10.255.255.3						
10.1.3.3	GigabitEthernet1	v2/S	1	30	1	10.1.3.3
10.255.255.3	Tunnel1	v2/P	0	30	1	

```

10.255.255.3
Leaf-R3#show tunnel endpoints
Tunnel1 running in MUDP/IP mode

Endpoint transport 10.255.255.4 Refcount 3 Base 0x77349A6A4CF8 Create Time
02:28:31
  overlay 10.255.255.4 port 4789 Refcount 2 Parent 0x77349A6A4CF8 Create Time
02:28:31
  connid 1
Endpoint transport 10.255.255.5 Refcount 3 Base 0x77349A6A4E88 Create Time
02:28:21
  overlay 10.255.255.5 port 4789 Refcount 2 Parent 0x77349A6A4E88 Create Time
02:28:21
  connid 1
Endpoint transport 239.6.7.8 Refcount 3 Base 0x77349A6A4B68 Create Time 00:01:25
  overlay 239.6.7.8 port 4789 Refcount 2 Parent 0x77349A6A4B68 Create Time
00:01:25
  connid 1

Leaf-R3#show ip mroute
IP Multicast Routing Table
Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
       L - Local, P - Pruned, R - RP-bit set, F - Register flag,
       T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
       X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
       U - URD, I - Received Source Specific Host Report,
       Z - Multicast Tunnel, z - MDT-data group sender,
       Y - Joined MDT-data group, y - Sending to MDT-data group,
       G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
       N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
       Q - Received BGP S-A Route, q - Sent BGP S-A Route,
       V - RD & Vector, v - Vector, p - PIM Joins on route,
       x - VxLAN group, c - PFP-SA cache created entry,
       * - determined by Assert, # - iif-starg configured on rpf intf,
       e - encap-helper tunnel flag, l - LISP decap ref count contributor
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                          t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:00:42/00:02:18, RP 10.255.255.1, flags: SJCx
  Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.1
  Outgoing interface list:
    Tunnel1, Forward/Sparse-Dense, 00:00:42/00:02:18, flags:

```

В MRIB мы видим starG-запись, которая говорит о наличии подписки на группу 239.6.7.8, в которой входящим интерфейсом выбран Gi1 (как интерфейс, смотрящий на RP), а исходящим – Tunnel1. Флаги, указанные в записи, говорят нам о следующем:

- **S** – **Sparse** – PIM в режиме Sparse
- **J** – **Join SPT** – по возможности будет переключение на SPT
- **C** – **Connected**
- **x** – **VxLAN group** – группа используется для VxLAN

Также стоит отметить о том, что R3 начал отправлять PIM-Join для группы 239.6.7.8 в сторону RP. На маршрутизаторах в дебагах это видно:

R3:

```
Leaf-R3#debug ip pim
PIM debugging is on
Leaf-R3#
*May 18 2025 17:35:47.551 MSK: PIM(0)[default]: Building Periodic (*,G) Join /
(S,G,RP-bit) Prune message for 239.6.7.8
*May 18 2025 17:35:47.551 MSK: PIM(0)[default]: Insert (*,239.6.7.8) join in nbr
10.1.3.1's queue
*May 18 2025 17:35:47.551 MSK: PIM(0)[default]: Building Join/Prune packet for nbr
10.1.3.1
*May 18 2025 17:35:47.551 MSK: PIM(0)[default]: Adding v2 (10.255.255.1/32,
239.6.7.8), WC-bit, RPT-bit, S-bit Join, RLOC_G: 0.0.0.0
*May 18 2025 17:35:47.551 MSK: PIM(0)[default]: Send v2 join/prune to 10.1.3.1
(GigabitEthernet1)
Leaf-R3#
```

R1:

```
Spine-R1#
*May 18 2025 17:35:47.552 MSK: PIM(0)[default]: Received v2 Join/Prune on
GigabitEthernet1 from 10.1.3.3, to us
*May 18 2025 17:35:47.552 MSK: PIM(0)[default]: Join-list: (*, 239.6.7.8), RPT-bit
set, WC-bit set, S-bit set
*May 18 2025 17:35:47.553 MSK: PIM(0)[default]: Re-check RP 10.255.255.1 into the
(*, 239.6.7.8) entry
*May 18 2025 17:35:47.554 MSK: PIM(0)[default]: Tunnel1 locked on
(0.0.0.0,239.6.7.8), lock TUN_MODE_PIM_DEC_IPV4_default refcnt 1, mdb
[0]:TUN_MODE_PIM_DEC_IPV4_default [1]:
*May 18 2025 17:35:47.554 MSK: PIM(0)[default]: Adding register decap tunnel
(Tunnel1) as accepting interface of (*, 239.6.7.8).
Spine-R1#
*May 18 2025 17:35:47.556 MSK: PIM(0)[default]: MIDB Add GigabitEthernet1/10.1.3.3
to (*, 239.6.7.8), Forward state, by PIM *G Join
Spine-R1#
```

MRIB на R1:

R1:

```
Spine-R1#show ip mroute
IP Multicast Routing Table
Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
L - Local, P - Pruned, R - RP-bit set, F - Register flag,
T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
```



```

    U - URD, I - Received Source Specific Host Report,
    Z - Multicast Tunnel, z - MDT-data group sender,
    Y - Joined MDT-data group, y - Sending to MDT-data group,
    G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
    N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
    Q - Received BGP S-A Route, q - Sent BGP S-A Route,
    V - RD & Vector, v - Vector, p - PIM Joins on route,
    x - VxLAN group, c - PFP-SA cache created entry,
    * - determined by Assert, # - iif-starg configured on rpf intf,
    e - encap-helper tunnel flag, l - LISP decap ref count contributor
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:12:52/00:03:26, RP 10.255.255.1, flags: S
Incoming interface: Null, RPF nbr 0.0.0.0
Outgoing interface list:
    GigabitEthernet1, Forward/Sparse, 00:12:51/00:03:26, flags:

```

Теперь добавим настройки сервиса на R3 и на R6:

R3:

```

interface GigabitEthernet4
 service instance 678 ethernet
 encapsulation dot1q 678
 exit
!
bridge-domain 678
 member Gi4 service-instance 678
 member vni 10002
!

```

R6:

```

interface Ethernet0/0.678
 encapsulation dot1q 678
 ip address 10.10.2.6 255.255.255.0
!

```

Когда мы добавили адрес на R6, то в дебагах увидим следующее:

R3:

```

Leaf-R3#
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Tunnel0 locked on
(10.255.255.3,239.6.7.8), lock TUN_MODE_PIM_ENC_IPV4_default refcnt 1, mdb

```

```
[0]:TUN_MODE_PIM_ENC_IPV4_default [1]:
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Adding register encap tunnel
(Tunnel0) as forwarding interface of (10.255.255.3, 239.6.7.8).
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Insert (10.255.255.3,239.6.7.8)
sgr prune in nbr 10.1.3.1's queue
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: previous iif and rpf_nbr is same
as new one, ignoring it
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Building Join/Prune packet for nbr
10.1.3.1
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Adding v2 (10.255.255.3/32,
239.6.7.8), RPT-bit, S-bit Prune, RLOC_G: 0.0.0.0
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Send v2 join/prune to 10.1.3.1
(GigabitEthernet1)
*May 18 2025 17:52:52.391 MSK: PIM(0)[default]: Received v2 Join/Prune on
GigabitEthernet1 from 10.1.3.1, to us
*May 18 2025 17:52:52.391 MSK: PIM(0)[default]: Join-list: (10.255.255.3/32,
239.6.7.8), S-bit set
*May 18 2025 17:52:52.391 MSK: PIM(0)[default]: MIDB Add GigabitEthernet1/10.1.3.1
to (10.255.255.3, 239.6.7.8), Forward state, by PIM SG Join
Leaf-R3#
*May 18 2025 17:52:52.391 MSK: PIM(0)[default]: Join to 0.0.0.0 on Loopback0 for
(10.255.255.3, 239.6.7.8), Ignored.
*May 18 2025 17:52:52.445 MSK: PIM(0)[default]: Received v2 Join/Prune on
GigabitEthernet1 from 10.1.3.1, to us
*May 18 2025 17:52:52.445 MSK: PIM(0)[default]: Join-list: (10.255.255.3/32,
239.6.7.8), S-bit set
*May 18 2025 17:52:52.445 MSK: PIM(0)[default]: MIDB Update
GigabitEthernet1/10.1.3.1 to (10.255.255.3, 239.6.7.8), Forward state, by PIM SG
Join
Leaf-R3#
```

R1:

```
Spine-R1#
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Received v2 Join/Prune on
GigabitEthernet1 from 10.1.3.3, to us
*May 18 2025 17:52:52.365 MSK: PIM(0)[default]: Prune-list: (10.255.255.3/32,
239.6.7.8) RPT-bit set
*May 18 2025 17:52:52.387 MSK: PIM(0)[default]: Received v2 Register on
GigabitEthernet1 from 10.1.3.3
*May 18 2025 17:52:52.387 MSK: for 10.255.255.3, group 239.6.7.8
*May 18 2025 17:52:52.388 MSK: PIM(0)[default]: Tunnel1 locked on
(10.255.255.3,239.6.7.8), lock TUN_MODE_PIM_DEC_IPV4_default refcnt 2, mdb
[0]:TUN_MODE_PIM_DEC_IPV4_default [1]:
*May 18 2025 17:52:52.388 MSK: PIM(0)[default]: Adding register decap tunnel
(Tunnel1) as accepting interface of (10.255.255.3, 239.6.7.8).
*May 18 2025 17:52:52.389 MSK: PIM(0)[default]: Insert (10.255.255.3,239.6.7.8)
join in nbr 10.1.3.3's queue
*May 18 2025 17:52:52.389 MSK: PIM(0)[default]: Received v2 Register on
GigabitEthernet1 from 10.1.3.3
*May 18 2025 17:52:52.389 MSK: for 10.255.255.3, group 239.6.7.8
*May 18 2025 17:52:52.389 MSK: PIM(0)[default]: Building Join/Prune packet for nbr
10.1.3.3
```

```

*May 18 2025 17:52:52.390 MSK: PIM(0)[default]: Adding v2 (10.255.255.3/32,
239.6.7.8), S-bit Join, RLOC_G: 0.0.0.0
Spine-R1#
*May 18 2025 17:52:52.390 MSK: PIM(0)[default]: Send v2 join/prune to 10.1.3.3
(GigabitEthernet1)
*May 18 2025 17:52:52.391 MSK: PIM(0)[default]: previous iif and rpf_nbr is same
as new one, ignoring it
*May 18 2025 17:52:52.443 MSK: PIM(0)[default]: Insert (10.255.255.3,239.6.7.8)
join in nbr 10.1.3.3's queue
*May 18 2025 17:52:52.444 MSK: PIM(0)[default]: Building Join/Prune packet for nbr
10.1.3.3
*May 18 2025 17:52:52.444 MSK: PIM(0)[default]: Adding v2 (10.255.255.3/32,
239.6.7.8), S-bit Join, RLOC_G: 0.0.0.0
*May 18 2025 17:52:52.444 MSK: PIM(0)[default]: Send v2 join/prune to 10.1.3.3
(GigabitEthernet1)
Spine-R1#show ip mroute

Spine-R1#show ip mroute
IP Multicast Routing Table
Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
L - Local, P - Pruned, R - RP-bit set, F - Register flag,
T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group, c - PFP-SA cache created entry,
* - determined by Assert, # - iif-starg configured on rpf intf,
e - encap-helper tunnel flag, l - LISP decap ref count contributor
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:17:15/stopped, RP 10.255.255.1, flags: S
Incoming interface: Null, RPF nbr 0.0.0.0
Outgoing interface list:
GigabitEthernet1, Forward/Sparse, 00:17:15/00:02:58, flags:

(10.255.255.3, 239.6.7.8), 00:00:10/00:02:49, flags: PX
Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.3
Outgoing interface list: Null

```

Что произошло? Когда мы добавили IP-адрес на R6, CE отправил Grarp, который, в свою очередь, дошел до VTEP R3. Т.к. ARP – это широковещательный кадр, он был упакован в мультикастовый VxLAN-пакет на группу 239.6.7.8, а он был упакован в register-туннель.

arp						
No.	Time	Source	Destination	Protocol	Length	Info
2086	2025-05-18 17:06:05,804450	50:00:00:01...	50:00:00:03:00:00	ARP	60	Who has 10.1.3.3? Tell 10.1.3.1
2087	2025-05-18 17:06:05,806395	50:00:00:03...	50:00:00:01:00:00	ARP	60	10.1.3.3 is at 50:00:00:03:00:00
4563	2025-05-18 17:52:52,405165	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	PIMv2	142	Register
4564	2025-05-18 17:52:52,405297	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	PIMv2	142	Register
4565	2025-05-18 17:52:52,408211	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	ARP	114	Gratuitous ARP for 10.10.2.6 (Reply)
4566	2025-05-18 17:52:52,408223	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	ARP	114	Gratuitous ARP for 10.10.2.6 (Reply)

```

> Frame 4564: 142 bytes on wire (1136 bits), 142 bytes captured (1136 bits) on interface -, id 0
> Ethernet II, Src: 50:00:00:03:00:00, Dst: 50:00:00:01:00:00
> Internet Protocol Version 4, Src: 10.1.3.3, Dst: 10.255.255.1
> Protocol Independent Multicast
> Internet Protocol Version 4, Src: 10.255.255.3, Dst: 239.6.7.8
> User Datagram Protocol, Src Port: 56271, Dst Port: 4789
> Virtual eXtensible Local Area Network
> Ethernet II, Src: aa:bb:cc:00:60:00, Dst: ff:ff:ff:ff:ff:ff
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 678
> Address Resolution Protocol (reply/gratuitous ARP)

```

На скриншоте видны все слои упаковки: ARP >> в dot1q с VID == 678 (т.к. мы не указывали rewrite-действие в EFP, метка влана сохранена) >> Ethernet >> VxLAN >> UDP >> IP (DST == 239.6.7.8) >> PIM Register >> IP (DST == 10.255.255.1) >> Ethernet.

После пары пакетов в register-Туннеле, они же были реплицированы в виде обычных мультикаст-пакетов:

4565	2025-05-18 17:52:52,408211	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	ARP	114	Gratuitous ARP for 10.10.2.6 (Reply)
4566	2025-05-18 17:52:52,408223	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	ARP	114	Gratuitous ARP for 10.10.2.6 (Reply)
4757	2025-05-18 17:56:25,690438	aa:bb:cc:00...	ff:ff:ff:ff:ff:ff	PIMv2	142	Register

```

> Frame 4566: 114 bytes on wire (912 bits), 114 bytes captured (912 bits) on interface -, id 0
> Ethernet II, Src: 50:00:00:01:00:00, Dst: 01:00:5e:06:07:08
> Internet Protocol Version 4, Src: 10.255.255.3, Dst: 239.6.7.8
> User Datagram Protocol, Src Port: 56271, Dst Port: 4789
> Virtual eXtensible Local Area Network
> Ethernet II, Src: aa:bb:cc:00:60:00, Dst: ff:ff:ff:ff:ff:ff
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 678
> Address Resolution Protocol (reply/gratuitous ARP)

```

Сейчас мы увидели, что действительно, BUM-трафик отправляется через multicast-рассылку. Давайте подключим CE-R7:

R4:

```

interface nve1
 member vni 10002 mcast-group 239.6.7.8
!
interface GigabitEthernet4
 service instance 678 ethernet
 encapsulation dot1q 678
 exit
!
bridge-domain 678
 member vni 10002
 member Gi4 service-instance 678
!

```

R7:

```
interface Ethernet0/0.678
 encapsulation dot1q 678
 ip address 10.10.2.7 255.255.255.0
!
```

В итоге мы получаем на RP такое:

R1:

```
Spine-R1#show ip mroute
IP Multicast Routing Table
Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
       L - Local, P - Pruned, R - RP-bit set, F - Register flag,
       T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
       X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
       U - URD, I - Received Source Specific Host Report,
       Z - Multicast Tunnel, z - MDT-data group sender,
       Y - Joined MDT-data group, y - Sending to MDT-data group,
       G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
       N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
       Q - Received BGP S-A Route, q - Sent BGP S-A Route,
       V - RD & Vector, v - Vector, p - PIM Joins on route,
       x - VxLAN group, c - PFP-SA cache created entry,
       * - determined by Assert, # - iif-starg configured on rpf intf,
       e - encap-helper tunnel flag, l - LISP decap ref count contributor
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                          t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:31:42/00:03:15, RP 10.255.255.1, flags: S
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    GigabitEthernet2, Forward/Sparse, 00:02:29/00:02:57, flags:
    GigabitEthernet1, Forward/Sparse, 00:31:42/00:03:15, flags:

(10.255.255.3, 239.6.7.8), 00:00:06/00:02:53, flags:
  Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.3
  Outgoing interface list:
    GigabitEthernet2, Forward/Sparse, 00:00:06/00:03:23, flags:

(10.255.255.4, 239.6.7.8), 00:00:23/00:02:36, flags:
  Incoming interface: GigabitEthernet2, RPF nbr 10.1.4.4
  Outgoing interface list:
    GigabitEthernet1, Forward/Sparse, 00:00:23/00:03:15, flags:

Spine-R1#
```

Стоит обратить внимание на то, что время работы обоих маршрутов небольшое – это связано с тем, что никаких BUM-пакетов на регулярной основе с обоих CE-устройств не приходит, поэтому периодически

эти маршруты исчезают (в реальной сети, разумеется, такое будет встречаться гораздо реже, т.к. трафик там есть постоянно). Включим еще один процесс OSPF на CE-R6 и CE-R7:

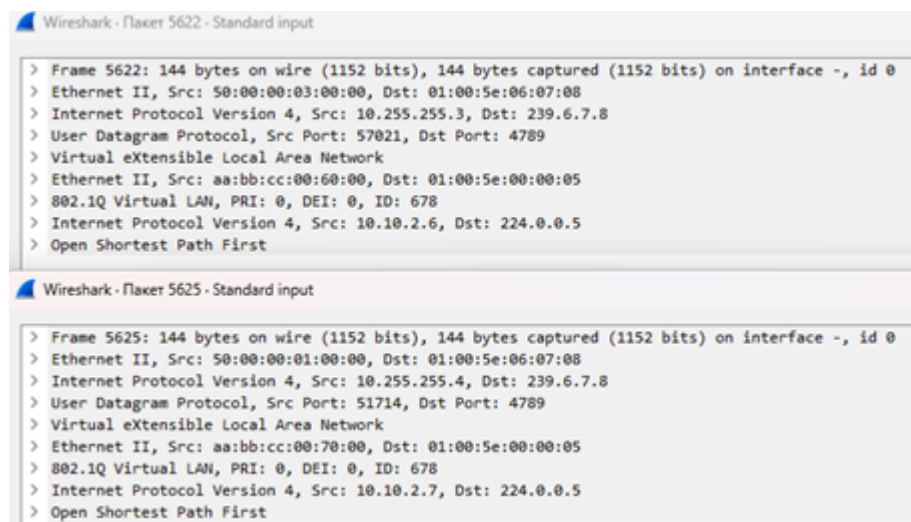
R6:

```
interface Loopback102
 ip address 102.0.0.6 255.255.255.255
 ip ospf 10002 area 0
!
interface Ethernet0/0.678
 ip ospf 10002 area 0
!
```

R7:

```
interface Loopback102
 ip address 102.0.0.7 255.255.255.255
 ip ospf 10002 area 0
!
interface Ethernet0/0.678
 ip ospf 10002 area 0
!
```

В дампе трафика (снимается все также на R3_Gi1) видим, что OSPF-пакеты ходят, упакованные в группу 239.6.7.8:



Давайте сейчас проверим обычное юникастовое взаимодействие:

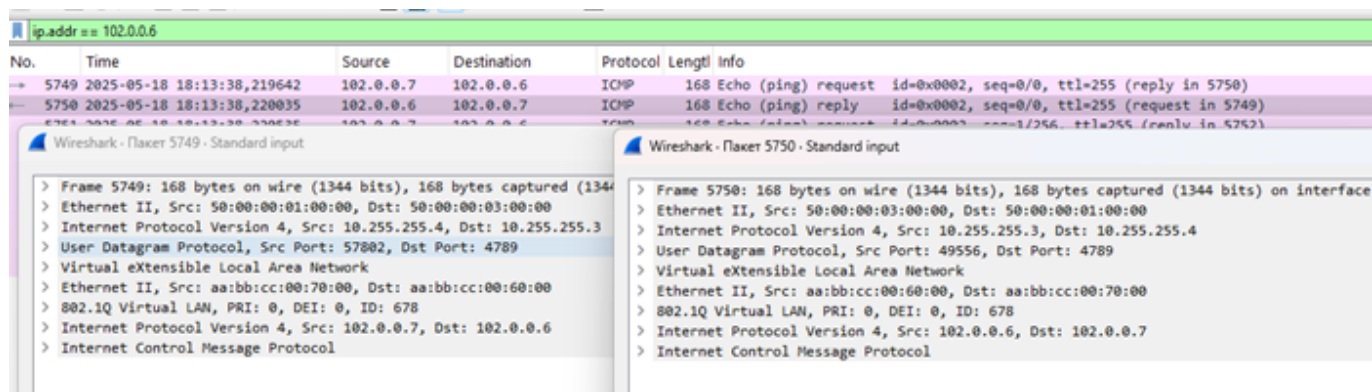
R7:

```
CE-R7#ping 102.0.0.6 source 102.0.0.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 102.0.0.6, timeout is 2 seconds:
Packet sent with a source address of 102.0.0.7
```

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

CE-R7#



Связность есть, а из дампа мы видим, что эти ICMP-запросы/ответы уже упакованы в юникастовые VxLAN-пакеты.

Давайте настроим оставшуюся пару VTEP-CE:

R5:

```
interface nve1
 member vni 10002 mcast-group 239.6.7.8
!
interface GigabitEthernet4
 service instance 678 ethernet
 encapsulation dot1q 678
 exit
!
bridge-domain 678
 member vni 10002
 member Gi4 service-instance 678
!
```

R8:

```
interface Loopback102
 ip address 102.0.0.8 255.255.255.255
 ip ospf 10002 area 0
!
interface GigabitEthernet1
 service instance 678 ethernet
 encapsulation dot1q 678
 rewrite ingress tag pop 1 symmetric
 exit
!
bridge-domain 678
 member Gi1 service-instance 678
!
```



```
interface BDI678
ip address 10.10.2.8 255.255.255.0
ip ospf 10002 area 0
no shutdown
!
```

Проверим связность:

R8:

```
CE-R8#ping 102.0.0.6 so Lo102
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 102.0.0.6, timeout is 2 seconds:
Packet sent with a source address of 102.0.0.8
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
CE-R8#ping 102.0.0.7 so Lo102
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 102.0.0.7, timeout is 2 seconds:
Packet sent with a source address of 102.0.0.8
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
CE-R8#
```

Посмотрим MRIB:

R1:

```
Spine-R1#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:54:42/00:03:24, RP 10.255.255.1, flags: S
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    GigabitEthernet3, Forward/Sparse, 00:10:54/00:03:24, flags:
    GigabitEthernet2, Forward/Sparse, 00:25:29/00:02:40, flags:
    GigabitEthernet1, Forward/Sparse, 00:54:42/00:02:56, flags:

(10.255.255.5, 239.6.7.8), 00:07:12/00:02:39, flags: T
  Incoming interface: GigabitEthernet3, RPF nbr 10.1.5.5
  Outgoing interface list:
    GigabitEthernet1, Forward/Sparse, 00:07:12/00:03:10, flags:
    GigabitEthernet2, Forward/Sparse, 00:07:12/00:03:13, flags:

(10.255.255.4, 239.6.7.8), 00:18:21/00:03:10, flags: T
  Incoming interface: GigabitEthernet2, RPF nbr 10.1.4.4
  Outgoing interface list:
    GigabitEthernet3, Forward/Sparse, 00:10:54/00:03:24, flags:
```



```
GigabitEthernet1, Forward/Sparse, 00:18:21/00:02:56, flags:
(10.255.255.3, 239.6.7.8), 00:19:11/00:02:37, flags: T
Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.3
Outgoing interface list:
GigabitEthernet3, Forward/Sparse, 00:10:54/00:03:24, flags:
GigabitEthernet2, Forward/Sparse, 00:19:11/00:02:59, flags:
```

R3:

```
Leaf-R3#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group
Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 02:02:46/stopped, RP 10.255.255.1, flags: SJCFx
Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.1
Outgoing interface list:
Tunnel1, Forward/Sparse-Dense, 02:02:46/00:00:14, flags:

(10.255.255.5, 239.6.7.8), 00:07:26/00:01:32, flags: JTx
Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.1
Outgoing interface list:
Tunnel1, Forward/Sparse-Dense, 00:07:26/00:01:33, flags:

(10.255.255.4, 239.6.7.8), 00:18:35/00:02:14, flags: JTx
Incoming interface: GigabitEthernet1, RPF nbr 10.1.3.1
Outgoing interface list:
Tunnel1, Forward/Sparse-Dense, 00:18:35/00:02:24, flags:

(10.255.255.3, 239.6.7.8), 00:19:26/00:02:22, flags: FTx
Incoming interface: Loopback0, RPF nbr 0.0.0.0
Outgoing interface list:
GigabitEthernet1, Forward/Sparse, 00:19:26/00:02:53, flags:
```

R4:

```
Leaf-R4#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group
Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:26:08/stopped, RP 10.255.255.1, flags: SJCFx
Incoming interface: GigabitEthernet1, RPF nbr 10.1.4.1
Outgoing interface list:
Tunnel1, Forward/Sparse-Dense, 00:26:08/00:00:53, flags:

(10.255.255.5, 239.6.7.8), 00:07:51/00:01:11, flags: JTx
```

```

Incoming interface: GigabitEthernet1, RPF nbr 10.1.4.1
Outgoing interface list:
  Tunnel1, Forward/Sparse-Dense, 00:07:51/00:01:08, flags:

(10.255.255.4, 239.6.7.8), 00:19:00/00:02:57, flags: FTx
Incoming interface: Loopback0, RPF nbr 0.0.0.0
Outgoing interface list:
  GigabitEthernet1, Forward/Sparse, 00:19:00/00:03:12, flags:

(10.255.255.3, 239.6.7.8), 00:19:50/00:01:30, flags: JTx
Incoming interface: GigabitEthernet1, RPF nbr 10.1.4.1
Outgoing interface list:
  Tunnel1, Forward/Sparse-Dense, 00:19:50/00:01:09, flags:

```

R5:

```

Leaf-R5#show ip mroute | b Outgoing
Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join
                        t - LISP transit group

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.6.7.8), 00:11:36/stopped, RP 10.255.255.1, flags: SJCFx
  Incoming interface: GigabitEthernet1, RPF nbr 10.1.5.1
  Outgoing interface list:
    Tunnel1, Forward/Sparse-Dense, 00:11:36/00:00:25, flags:

(10.255.255.5, 239.6.7.8), 00:07:53/00:01:57, flags: FTx
  Incoming interface: Loopback0, RPF nbr 0.0.0.0
  Outgoing interface list:
    GigabitEthernet1, Forward/Sparse, 00:07:53/00:02:31, flags:

(10.255.255.3, 239.6.7.8), 00:11:28/00:01:22, flags: JTx
  Incoming interface: GigabitEthernet1, RPF nbr 10.1.5.1
  Outgoing interface list:
    Tunnel1, Forward/Sparse-Dense, 00:11:28/00:00:31, flags:

(10.255.255.4, 239.6.7.8), 00:11:33/00:01:19, flags: JTx
  Incoming interface: GigabitEthernet1, RPF nbr 10.1.5.1
  Outgoing interface list:
    Tunnel1, Forward/Sparse-Dense, 00:11:33/00:00:25, flags:

```

Как мы видим, связность есть (как для юникаста, так и для BUM), в MRIB есть все необходимые записи (как starg, так и sg). На этом создание L2VNI в режиме Flood and Learn, с Multicast-group для BUM-трафика завершено.

Примененная конфигурация

Примененная конфигурация - L2VNI Ingress Replication solution config

Leaf-R3:

```
!  
interface GigabitEthernet4  
  service instance 6 ethernet  
  encapsulation dot1q 6  
  rewrite ingress tag pop 1 symmetric  
!  
!  
interface nve1  
  source-interface Loopback0  
  member vni 10001  
  ingress-replication 10.255.255.4  
  ingress-replication 10.255.255.5  
!  
!  
bridge-domain 1001  
  member vni 10001  
  member GigabitEthernet4 service-instance 6  
!
```

Leaf-R4:

```
!  
interface GigabitEthernet4  
  service instance 7 ethernet  
  encapsulation dot1q 7  
  rewrite ingress tag pop 1 symmetric  
!  
!  
interface nve1  
  source-interface Loopback0  
  member vni 10001  
  ingress-replication 10.255.255.3  
  ingress-replication 10.255.255.5  
!  
!  
bridge-domain 1001  
  member vni 10001  
  member GigabitEthernet4 service-instance 7  
!
```

Leaf-R5:

```
!  
interface GigabitEthernet4  
  service instance 8 ethernet  
  encapsulation dot1q 8  
  rewrite ingress tag pop 1 symmetric  
!  
!
```

```
interface nve1
  source-interface Loopback0
  member vni 10001
    ingress-replication 10.255.255.3
    ingress-replication 10.255.255.4
  !
!
bridge-domain 1001
  member vni 10001
  member GigabitEthernet4 service-instance 8
!
```

CE-R6:

```
!
interface Ethernet0/0.6
  description VNI_10001
  encapsulation dot1Q 6
  ip address 10.10.1.6 255.255.255.0
  ip ospf 10001 area 0
!
```

CE-R7:

```
!
interface Ethernet0/0.7
  description VNI_10001
  encapsulation dot1Q 7
  ip address 10.10.1.7 255.255.255.0
  ip ospf 10001 area 0
!
```

CE-R8:

```
!
interface GigabitEthernet1
  service instance 8 ethernet
  encapsulation dot1q 8
  rewrite ingress tag pop 1 symmetric
!
!
bridge-domain 8
  member GigabitEthernet1 service-instance 8
!
interface BDI8
  description VNI_10001
```

```
ip address 10.10.1.8 255.255.255.0
ip ospf 10001 area 0
```

Примененная конфигурация - L2VNI Multicast Group

Leaf-R3 / Leaf-R4 / Leaf-R5:

```
!
interface GigabitEthernet4
  service instance 678 ethernet
  encapsulation dot1q 678
!
!
interface nve1
  source-interface Loopback0
  member vni 10002 mcast-group 239.6.7.8
!
bridge-domain 678
  member vni 10002
  member GigabitEthernet4 service-instance 678
!
```

CE-R6:

```
!
interface Ethernet0/0.678
  description VNI_10002
  encapsulation dot1Q 678
  ip address 10.10.2.6 255.255.255.0
  ip ospf 10002 area 0
!
```

CE-R7:

```
!
interface Ethernet0/0.678
  description VNI_10002
  encapsulation dot1Q 678
  ip address 10.10.2.7 255.255.255.0
  ip ospf 10002 area 0
!
```

CE-R8:

```
!
interface GigabitEthernet1
```

```
service instance 678 ethernet
  encapsulation dot1q 678
  rewrite ingress tag pop 1 symmetric
!
!
bridge-domain 678
  member GigabitEthernet1 service-instance 678
!
!
interface BDI678
  ip address 10.10.2.8 255.255.255.0
  ip ospf 10002 area 0
!
```